Is Simple Enough? Reproducing and Stress-Testing "SGC ≈ GCN"

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Task and motivation

Let $\hat{A} = \tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$ with $\tilde{A} = A + I$, $\tilde{D} = D + I$. The two-layer GCN (Kipf & Welling, 2017) stacks the layer $H^{(k+1)} = \sigma(\hat{A}H^{(k)}W^{(k)})$ twice, $Z = \hat{A}\,\mathrm{ReLU}(\hat{A}XW_1)\,W_2$. Wu et al. (2019) delete the ReLU, so the model collapses to $Z = \hat{A}^K X\,\Theta$ with $\Theta = W_1W_2$: a fixed low-pass filter with frequency response $(1-\tilde{\lambda})^K$, followed by multinomial logistic regression on $\bar{X} = \hat{A}^K X$ (SGC). They report that SGC matches the GCN on semi-supervised node classification while training much faster. We will (i) reproduce this comparison from scratch and log every underspecified detail, then ask whether the claim holds under stricter measurement: (ii) is the gap distinguishable from *split noise*, (iii) does it survive *equal tuning budgets*, and (iv) does it survive on *heterophilous* graphs, where low-pass filtering should hurt? SGC is often recommended as a "first model to try"; we want to know when that advice is safe.

Datasets

Homophilous: Cora, Citeseer, Pubmed with the public Planetoid split ($|\mathcal{V}_{\text{train}}|$ = 20 per class, 500 val, 1,000 test), plus 100 random splits of the same sizes. **Heterophilous:** Roman-empire (≈22.7k nodes, 18 classes) and Amazon-ratings (≈24.5k nodes, 5 classes) from Platonov et al. (2023), with their 10 fixed splits. They are large enough for reliable comparisons, avoid the duplicate-node leakage of older heterophily benchmarks, and use accuracy like the citation graphs. For every graph we report edge homophily $h = |\{(u,v) \in E : y_u = y_v\}| \,/\, |E|$. All datasets load through PyTorch Geometric; everything runs on CPU.

Reproduction target (test acc., %)	Cora	Citeseer	Pubmed	Source
GCN (2 layers)	81.5	70.3	79.0	Kipf & Welling, Table 2 (mean of 100 inits)
SGC (K=2)	81.0 ± 0.0	71.9 ± 0.1	78.9 ± 0.0	Wu et al., Table 2 (10 runs)

Methods

GCN: two layers, $W_1 \in R^{d\times 16}$, $W_2 \in R^{16\times C}$, dropout 0.5, weight decay $5{\cdot}10^{-4}$ on W_1, Adam lr 0.01, $\leq$200 epochs, early stopping window 10, cross-entropy over $\mathcal{V}_{\text{train}}$. **SGC:** precompute $\bar{X} = \hat{A}^K X$ once, then train $\Theta \in R^{d\times C}$ with the same loss plus tuned weight decay (Adam lr 0.2, 100 epochs). Both are written by us in PyTorch, with no reference code reused. **Baselines:** an MLP on X and SGC with K=0 (logistic regression on X), so the benefit of the graph is measured, not assumed. **Ablation:** $K \in \{0,1,2,4,8\}$ for SGC and 1–4 layers for the GCN on every dataset.

Evaluation plan (fixed before any results)

1. **Reproduction.** Mean ± std over 10 seeds on the Planetoid split; a number counts as reproduced if it lies within 1 point of the paper. Every deviation is recorded in a discrepancy log (e.g. SGC's weight-decay search range, whether X is row-normalized before computing $\hat{A}^K X$, dropout placement, what early stopping monitors, 100 vs. 10 runs, std vs. standard error).
2. **Split variance and significance.** On 100 random splits s, both models see identical splits. We report the paired difference $\delta_s = \mathrm{acc}_{\mathrm{GCN}}(s) - \mathrm{acc}_{\mathrm{SGC}}(s)$ with a 95% bootstrap CI and a Wilcoxon signed-rank test, and split the GCN's variance into split and initialization parts (SGC's convex loss is nearly seed-independent). Success: a clear statement of whether the reported 1.6-point Citeseer gap exceeds split noise.
3. **Fair tuning.** Kipf tuned the GCN on Cora only and reused it elsewhere; Wu tuned SGC per dataset with 60 hyperopt trials. We compare GCN (paper defaults), GCN (60 trials per dataset over lr, weight decay, dropout, hidden width), and SGC (60 trials over weight decay). Success: whether SGC's Citeseer advantage remains once tuning is equal.
4. **Heterophily.** Same models and ablation on Roman-empire and Amazon-ratings. We plot accuracy minus MLP accuracy against h across all five graphs. Hypothesis: as h falls, the label signal moves to high $\tilde{\lambda}$, which $(1-\tilde{\lambda})^K$ suppresses, so both models drop below the MLP and SGC degrades faster as K grows. A negative result (SGC ≈ GCN everywhere) is also informative and will be reported as such.

Main risk and fallback

Risk: with only two heterophilous graphs, which also use a higher label rate (50/25/25) than Planetoid, a trend in h could be confounded by other dataset differences. **Fallback:** subsample $\mathcal{V}_{\text{train}}$ to 20 labels per class to match the citation setup, and add synthetic contextual stochastic block model graphs that sweep h from 0.1 to 0.9 with everything else fixed. **Milestones:** reproduction and split-variance study by the midway check (Nov 12); fair tuning, heterophily and report by Dec 3.

References. Kipf & Welling, *Semi-Supervised Classification with Graph Convolutional Networks*, ICLR 2017. Wu et al., *Simplifying Graph Convolutional Networks*, ICML 2019. Yang et al., *Revisiting Semi-Supervised Learning with Graph Embeddings*, ICML 2016. Platonov et al., *A Critical Look at the Evaluation of GNNs under Heterophily: Are We Really Making Progress?*, ICLR 2023.